

DEFECT ANALYSIS

Test 1: Valid Permit Verification Purpose: Confirm fix allows valid permits to work

Scenario: - Permit expires: May 30, 2026 at 10:00 PM

Current time: May 15, 2026 at 2:00 PM - Difference: 15 days remaining

Expected: System shows " PERMIT VALID - Parking Allowed"

Test 2: Expired Permit Verification Purpose: Confirm expired permits are still rejected

Scenario: - Permit expires: May 1, 2026 at 10:00 PM - Current time: May 15, 2026 at 2:00

PM - Permit is clearly past expiration

Expected: System shows " PERMIT EXPIRED - Parking Denied"

Test 3: Boundary Case (Today's Expiration)

Purpose: Confirm edge case works Scenario: - Permit expires: Today (May 15, 2026) at 11:59

PM - Current time: May 15, 2026 at 2:00 PM - Still has 10 hours before expiration

Expected: System shows " PERMIT VALID - Parking Allowed"

Defect Category: Timezone/Date Comparison Logic Error

Severity: CRITICAL

Reason for Severity: Valid permits are incorrectly rejected, preventing legitimate parking.

Students are unable to park even with valid permits and are being falsely fined. This impacts the core functionality of the system.

Priority: CRITICAL

Reason for Priority: Students are affected immediately during peak parking times. The university faces legal/financial risk from false violations. This requires immediate fix (within hours).

Root Cause: The system stores permit expiration dates in UTC timezone but compares them against local system time without converting timezones. When a student with a permit

expiring at 10:00 PM Pacific Time is verified, the system sees 5:00 AM UTC the next day, incorrectly marking it as expired.

Regression Test: • Test 1: Valid permit (expires in 15 days) → Should show as VALID • Test 2: Expired permit (expired 2 weeks ago) → Should show as EXPIRED • Test 3: Permit expiring today at 11:59 PM → Should show as VALID • Test 4: Same permit checked across different timezone servers → All should show as VALID